1 Point-set Topology and Separation Axioms

In what follows we set $(X, \mathcal{T})$ as a topological space. We skipped some definitions and denote the interior of a set $A \subseteq X$ to be $A^\circ := \bigcup\{V : V \text{ is a open set containing } A\}$.

Theorem 1. $(A^\circ)^c = \overline{A^c}, (A^c)^\circ = (\overline{A})^c$. *This follows from de Morgan's Law.*

Definition 1. A set $D \subseteq X$ is **dense** if $\overline{D} = X$, it is **nowhere dense** if $(\overline{A})^\circ = \emptyset$

Definition 2. X is **separable** if there is a countable dense set.

Definition 3. A/an (open) **neighbourhood base at** $x \in X$ is a collection of (open) sets $N_x \subset \mathcal{T}$ such that:

- $x \in V$ for all $V \in N_x$, and
- If U open, $x \in U$ (we sometimes write this as *if U is in the neighbourhood of x*), then there exists $V \in N_x$ such that $V \subseteq U$.

A **base** of the topology $\mathcal{T}$ is the set $\mathcal{B} := \bigcup_{x \in X} N_x$. It follows that, given a base $\mathcal{B}$, the set $M_x := \{N : N \in \mathcal{B}, x \in N\}$ is a neighbourhood base at x.

Definition 4. The **topology generated by** any set $E \subseteq \mathcal{P}(X)$ is the set

$$\mathcal{T}(E) := \left\{ \bigcap_{(X,J) \text{is a topological space, } E \subset J} J \right\}$$

i.e. the smallest topology such that all the sets in E is open, yet equivalently, $\mathcal{T}(E)$ is arbitrary union of finite intersections of sets in E.
E is also called the **subbase** of $\mathcal{T}(E)$ (usually when $\mathcal{T}(E)$ is given).

Theorem 2. *The following are equivalent:*

1. *$\mathcal{B}$ is a base of $\mathcal{T}$.*
2. *Every non-empty open set is a union of set in $\mathcal{B}$.*
3. - *For all $x \in X$, there is $A \in \mathcal{B}$ such that $x \in A$*
   - *If $U, V \in \mathcal{B}$ and $x \in U \cap V$, then there exists $W \in \mathcal{B}$ such that $x \in W$ and $W \subseteq U \cap V$.*

   From here, finite intersection of elements in E forms a base of $\mathcal{T}(E)$.

Definition 5. A topological space is **first countable** if every point x has a countable neighbourhood base N_x at that point. It is **second countable** if there is a countable base.

Second countable $\Rightarrow$ First countable, but converse don't hold by uncountable discrete (two open sets) space. Also, if X is first countable, a neighbourhood space can be WLOG considered nested inside one another.

Definition 6. A sequence in X, denotes by x_j, **converges to** $x \in X$ if for any open U containing x, there is a $N \in \mathbb{N}$ such that for all $n \geqslant N, x_n \in U$

This is essentially a restatement of the ε-δ condition, and we also have the limit point characterization as in metric spaces:

Theorem 3. *If X is first countable, then $x \in \overline{A} \Leftrightarrow$ there is a sequence in A converging to x.*

Definition 7 (Separation Axioms)**.** A space is

- T_0: $x \neq y \Rightarrow$ There exists open set U such that $x \in U, y \notin U$ or $x \notin U, y \in U$.
- T_1: $x \neq y \Rightarrow$ There exists open set U such that $x \in U, y \notin U$.
- T_2: $x \neq y \Rightarrow$ There exists open set U, V such that $x \in U, y \in V, U \cap V = \emptyset$.
- T_3: X is T_1, and any closed set $A \subseteq X, x \in A^c$, there exists open set U, V such that $A \subseteq U, x \in V, U \cap V = \emptyset$.
- T_4: X is T_1, and any closed set $A, B \subseteq X, A \cap B = \emptyset$, there exists open set U, V such that $A \subseteq U, B \subseteq V, U \cap V = \emptyset$.

T_2, T_3, T_4 are called **Hausdorff, Regular, Normal** spaces respectively.

Note that T_0, T_1 are different because the open set U is specified. Also, in any T_1 spaces, $\{x\}$ is a closed set. Also, sometimes regular and normal spaces are referred without the "X is T_1" condition.

2 Continuous Functions and its Extensions

Definition 8. A function $f : X \to Y$ is **continuous at** x if for all V containing $f(x)$, there is open $U \subseteq X$ such that $U \subseteq f^{-1}(V)$.

This is equivalent to the continuous definition in topological spaces. Moreover, it suffices to check at the base:

Theorem 4. *If $\mathcal{T}(\mathcal{E})$ is a the topology of Y, then $f : X \to Y$ is continuous $\Leftrightarrow$ $f^{-1}(V)$ is open for all open $V \in \mathcal{E}$.*

Now, motivated by $\mathbb{R}^n$, we construct products of topological spaces from the projections:

Definition 9. If X is a SET, $(Y_\alpha, \mathcal{S}_\alpha)$ be a collection of topological spaces with an index set A. $f_\alpha : X \to Y_\alpha$ are set-functions: note that the discrete topology always makes f_α continuous.
The **weak topology on** X **with respect to** $\{f_\alpha\}_{\alpha \in A}$ is the "smallest" (weakest) topology that makes all the f_α continuous.

Definition 10. Given a set X, an index set A, we can make the Cartesian product $X^A := \prod_{\alpha \in A} X_\alpha$, the **product topology on** X^A is the weak topology with respect to the projections.

Theorem 5. *This set forms a basis of the product topology on X^A:*

$$\left\{ \prod_{\alpha \in A} V_\alpha : V_\alpha = U_\alpha, \text{ where } U_\alpha \text{ is some open set in } X \text{ for finitely many } \alpha, \text{ otherwise } V_\alpha = X_\alpha \right\}$$

X^A can be realized as $\{f | f : A \to X\}$.

Theorem 6 (Weak Topology is Pointwise Convergence)**.** *The following are equivalent:*

- *A sequence in X^A, denoted by $\{f_n\}_{n \in \mathbb{N}}$ is convergent to $f \in X^A$ in the product topology.*
- *For all $\alpha \in A$, $f_n(\alpha) \in X$ converge to $f(\alpha) \in X$ in the topology of X.*

We now turn to continuous functions with real outputs. Note that the same argument applies to $\mathbb{C}$ too.
Notation: we use $B(X, \mathbb{R})$ to denote the bounded functions from X to $\mathbb{R}$, $C(X, \mathbb{R})$ as the continuous functions, and bounded and continuous, $BC(X, \mathbb{R})$. With the above remark, we might as well drop the $\mathbb{R}$ too if we like.

Theorem 7 (Urysohn's Lemma)**.** *If X is T_4 and $A, B \subseteq X$ are disjoint and closed, then there is $f \in C(X, [0, 1])$ such that $f \equiv 0$ on A, $f \equiv 1$ on B.*

Theorem 8 (Tietze's Extension Lemma)**.** *If $A \subseteq X$, A closed, X is T_4, then given any $f \in C(A, [0, 1])$, there is $F \in C(X, [0, 1])$ such that $F|_A = f$.*

T_4 is kinda too strong for this to work, we can develop a more general theory following:

3 Other Nice Spaces, Topology on $C(X)$

Definition 11. A topological space X is **LCH** if

1. X is Hausdroff.
2. For any $x \in X$, there is open set $x \in U$ such that $\overline{U}$ is compact.

Theorem 9. *Some facts about compact and LCH spaces:*

- *Any Compact Hausdroff space is T_4.*
- *Closed subset of a compact set is compact.*
- *If X is LCH, any open neighbourhood U of $x \in X$ has a precompact (open) set V sit inside U.*
- *If X is LCH, E closed $\Leftrightarrow$ $E \cap K$ is closed for all compact K.*

Theorem 10 (Urysohn's Lemma, LCH Version)**.** *If X is LCH, K compact, U open, such that $K \subset U \subseteq X$, then there is $f \in C(X, [0,1])$ such that $f \equiv 0$ outside a compact subset of U, $f \equiv 1$ on K.*

Theorem 11 (Tietze's Extension Lemma, LCH Version)**.** *If X is LCH, K compact, U open, such that $K \subset U \subseteq X$, then given any $f \in C(K, [0,1])$, there is $F \in C(K, [0,1])$ such that $F|_K = f$ and F vanishes outside a compact set containing K.*

Definition 12. A topological space X is **σ-compact** if it is a countable union of compact sets in the topology.

Theorem 12. *If X is LCH and σ-compact, then there is open set collection $\{U_i\}_{i=1}^{\infty}$ such that for all n,*

$$U_n \text{ is precompact, } \bigcup_{i=1}^{\infty} U_i = X, \text{ and } \overline{U_n} \subseteq U_{n+1}.$$

Definition 13. The **support** of $f \in C(X)$, is the set

$$\operatorname{supp}(f) := \overline{\{x : f(x) \neq 0\}}$$

If $\operatorname{supp}(f)$ is compact, we say f is **compactly supported**, or we write $f \in C_c(X)$.

Definition 14. If for all $\varepsilon > 0$, the set $\{x : |f(x)| \geqslant \varepsilon\}$ is compact, we say f **vanishes at infinity** and write $f \in C_0(X)$.

Theorem 13. *If X is LCH, $C(X)$ is given the uniform metric topology, then $\overline{C_c(X)} = C_0(X)$.*

Definition 15. $\|f\|_u = \sup_x \|f(x)\|$ is called the **uniform norm** on $B(X)$, it makes $B(X)$ a metric space using $\rho(f,g) = \|f - g\|_u$.

Given a topological space X, what are the topology on $C(X)$? It turns out that there are 3 we are concerned with. To specify a topology, we only need to specify a neighbourhood basis around a point $f \in C(X)$.

- (Product / Weak Topology / Pointwise convergence.) We call this topology $\mathbb{T}_{pt}$. Note that by Theorem 6, weak and pointwise are the same topology. Hence, a local basis is:

$$\mathcal{B}(f) = \left\{ g \in C(X) : |f(x_i) - g(x_i)| < \frac{1}{m};\ x_1 \cdots x_n \in X, m \in \mathbb{N} \right\}$$

- (NEW!) (Uniform on Compact subset.) We call this topology $\mathbb{T}_{u.comp}$. We have a local basis:

$$\mathcal{B}(f) = \left\{ g \in C(X) : \sup_{x \in K} |f(x) - g(x)| < \frac{1}{m};\ K \text{ is any compact set} \subseteq X, m \in \mathbb{N} \right\}$$

- (Uniform Topology.) We call this topology $\mathbb{T}_{unif}$. This is just the metric topology on $\|.\|_u$. We have a local basis:

$$\mathcal{B}(f) = \left\{ g \in C(X) : \sup_{x \in X} |f(x) - g(x)| < \frac{1}{m};\ m \in \mathbb{N} \right\}$$

Then it is easy to see on $C(X)$, $\mathbb{T}_{pt} \subseteq \mathbb{T}_{u.comp} \subseteq \mathbb{T}_{unif}$.

Theorem 14. *Both $B(X), BC(X)$ is a complete space under ρ.*

Theorem 15. *If X is LCH, then $C(X)$ is closed in $\mathbb{T}_{u.comp}$. If further, X is σ-compact, then $\mathbb{T}_{u.comp}$ is first countable.*

In particular, $C(X)$ is closed on $\mathbb{T}_{unif}$, not necessarily on $\mathbb{T}_{pt}$.

4 Some Topology Theorems

Theorem 16 (Tychonoff Product)**.** *Any family of compact topological space $\{X_\alpha\}_{\alpha\in A}$, then their product is compact in the product topology. (Requires the axiom of choice.)*

The proof of this theorem uses the language of nets, which generalizes sequence. Here is a brief introduction to it:

Definition 16. A **directed set** $(A, \precsim)$ is a pair of set and a binary relation that satisfies: for any $\alpha, \beta, \gamma \in A$:

- (Reflexive and Transitive.) $\alpha \precsim \alpha; \alpha \precsim \beta, \beta \precsim \gamma \Rightarrow \alpha \precsim \gamma$
- (Maximal existence.) There exists $\gamma \in A$ such that $\alpha \precsim \gamma, \beta \precsim \gamma$.

A function from a directed set $A \to X$ is called a **net**, and is denoted by $\langle x_\alpha \rangle_{\alpha\in A}$

One of the key example is **directed set by reverse inclusion**, if X is a topological space, then the set $(\mathcal{N}_x, \supseteq)$ where $\mathcal{N}_x$ is an open set containing a marked point $x \in X$.

Note that the notion of convergence $\langle x_\alpha \rangle \to x_0$ can be similarly defined following the definition of sequences.

Theorem 17. *A function is continuous $\Leftrightarrow$ there exists a net which demonstrate sequential continuity.*

This skips the first countability for normal sequences. Note that the rightward proof is done by picking the reverse inclusion directed set.

Definition 17. A **subnet** of a net $\langle x_\alpha \rangle_{\alpha\in A}$, is a net $\langle y_\beta \rangle_{\beta\in B}$ with a specification of index (i.e. a function $f : B \to A$ denoted by $f(\beta) = \alpha_\beta$)

- $y_\beta = x_{\alpha_\beta}$, and;
- For all $\alpha \in A$, there is $\beta_0 \in B$ such that $\alpha_\beta \succsim \alpha_0$ for all $\beta \succsim \beta_0$.

Theorem 18 (Heine-Borel Property)**.** *Let X be a topological space, then:*

X is compact $\Leftrightarrow$ Every net in X has a "limit point" in X $\Leftrightarrow$ Every net have a subnet that converge to a point in X.

Definition 18. A collection $\mathcal{F} \subset C(X)$ is **equicontinuous** if for all $\varepsilon > 0, x \in X$, there is open set U_x on X such that $|f(y) - f(x)| < \varepsilon$ for all $f \in \mathcal{F}, y \in U_x$.

Definition 19. A collection $\mathcal{F} \subset C(X)$ is **pointwise bounded** if for every $x \in X$, the set $\mathcal{F}_x := \{f(x) : f \in \mathcal{F}\}$ is bounded.

Theorem 19 (Arzela-Ascoli)**.** *If X is compact and Hausdorff, $\mathcal{F} \subset C(X)$ is equicontinuous and pointwise bounded, then $\mathcal{F}$ is a totally bounded set in the metric topology of uniform norm, i.e. $\overline{\mathcal{F}}$ is compact.*

Definition 20. An **algebra** $\mathcal{A} \subset C(X, \mathbb{R})$ if $\mathcal{A}$ is a vector space over $\mathbb{R}$, and $\mathcal{A}$ is closed under pointwise multiplication.

An algebra **separates points** if for every $x, y \in X$ that are distinct, there is $g \in \mathcal{A}$ such that $g(x) \neq g(y)$.

Theorem 20 (Stone-Weierstrass)**.** *If X is a compact Hausdroff space, $\mathcal{A}$ is a closed subalgebra of $C(X, \mathbb{R})$ that separates points, then either*

- *$\mathcal{A} = C(X, \mathbb{R})$, or;*
- *There is $x_0 \in \mathbb{R}$ such that $\mathcal{A} = \{f(x) : f(x_0) = 0\}$.*

The important theorems have a generalization to σ-compact, LCH space, usually by the diagonal argument.

5 Some Functional Analysis

In what follows, V always denote a vector space over $\mathbb{R}$ or $\mathbb{C}$, where $\mathbb{R}, \mathbb{C}$ are always given the euclidean standard topology.

Definition 21. A function $V \to [0, \infty$ is a **seminorm** if it satisfies homogenety and the triangle inequality. It is a **norm** if $\|x\| = 0 \Rightarrow x = 0$, note that a norm always induces a metric (topology) by $\rho(x, y) = \|x - y\|$.

A **Banach space** is a complete normed vector space.

Theorem 21. *A vector space is Banach $\Leftrightarrow$ every aboslutely convergent series converges.*

In Analysis, we are interested in continuous (or "nice") maps between vector space. It turns out that for linear maps there is a very nice result.

Definition 22. A linear map $T : V \to W$ is **bounded** if there is $c < \infty$ such that $\|Tx\| \leqslant c\|x\|$ for all $x \in V$.

We also denote $\mathcal{L}(V, W) := \{T : T \text{ is a bounded linear map } V \to W\}$. We further denote $\mathcal{L}(V, \mathbb{R})$ or $L(V, \mathbb{C})$ as the **dual** V^*, the definition depends whether V is a vector space over $\mathbb{R}$ or $\mathbb{C}$.

Theorem 22. *Some collection of results: let $T \in \mathcal{L}(V, W)$:*

- *(See Homework 16.1) The expression*

$$\|T\| = \sup\{\|Tx\| : \|x\| = 1\} = \sup\left\{\frac{\|Tx\|}{\|x\|} : x \neq 0\right\} = \inf\{C : \|Tx\| \leqslant C\|x\| \text{ for all } x\}.$$

  *is a norm on the vector space $\mathcal{L}(V, W)$. This is called the **operator norm** sometimes.*

- *If Y is Banach, so is $L(X, Y)$, so V^* is automatically a Banach space.*

- *The following are equivalent:*

$$T \text{ is uniformly continuous, } T \text{ is continuous, } T \text{ is continuous at } 0, \ T \text{ is bounded.}$$

The next idea is the Hahn-Banach theorem. The main motivavtion is kind of copying the Urysohn lemma (So, the following discussion only works for vector space over $\mathbb{R}$), of asking is there a way to have a functional $f \in V^*$ such that $f \equiv 0$ on a subspace $\mathcal{M} \leqslant V$ and $f \equiv 1$ on the remaining. Looking at finite dimensional space this seems impossible, with only the projection function (defined by dot product) satisfy the idea of "the function's output grows as we go "away" from $\mathcal{M}$. We want to mimic this idea in the infinite dimensional space.

Definition 23. A **sublinear function(al)** p is a function $V \to \mathbb{R}$ such that the triangle inequality is satisfied, and *positive homogeneity* holds: $p(rx) = rp(x)$ for all $r \geqslant 0, x \in X$. (Note: another equivalent definition of positive homogeneity is *non-negative homogeneity*, which requires $p(rx) \leqslant rp(x)$ for all $r > 0, x \in X$)

Theorem 23 (Hahn-Banach)**.** *If V is a vector space over $\mathbb{R}$, $\mathcal{M}$ a subspace of V, $f : M \to \mathbb{R}$ linear, p a sublinear functional, then there is $F \in V^*$ such that $F(x) \leqslant p(x)$ for all $x \in V$, and $F|_{\mathcal{M}} = f$.*

Theorem 24 (Consequences of Hahn-Banach)**.** *If X is a normed vector space, then*

- *If $\mathcal{M}$ is a closed subspace of X and $x \in X$ $\mathcal{M}$, there exists $f \in X^*$ such that $f(x) \neq 0$ and $f|_{\mathcal{M}} = 0$. In fact, if $\delta = \inf_{y \in \mathcal{M}} \|x - y\|$, f can be taken to satisfy $\|f\| = 1$ and $f(x) = \delta$.*

- *If $x \neq 0 \in X$, there exists $f \in X^*$ such that $\|f\| = 1$ and $f(x) = \|x\|$.*

- *The bounded linear functionals on X separate points.*

- *If $x \in X$, $\widehat{x} : X^* \to \mathbb{C}$ by $\widehat{x}(f) = f(x)$. Then the **natural isomorphism** $x \mapsto \widehat{x}$ is a linear isometry from X into X^{**}.*

Note that the natural isomorphism is not necessarily a surjection, but it is always an injection.

Definition 24. A vector space X is **reflexive** if $X \cong X^{**}$.

6 L^p spaces

The proof is illuminating in L^p spaces, so we will present the main definitions and theorems first, which maybe less interesting, then provide the relevant proof and highlight the important ingredient.

In what follows $(X, \mathcal{M}, \mu)$ is a measure space. f, g are $X \to \mathbb{R}$ functions. Let $p > 0$:

Definition 25. If $p < \infty$, define:

$$L^p := \left\{ f : \int_X |f|^p \, d\mu < \infty \right\} / (\text{Almost everywhere equivalence relation})$$

Definition 26. The **essential supremum** of a function f is defined as:

$$\operatorname{ess\,sup}(f) = \inf_{x \in X} \{\alpha : \mu(\{y : |f(y)| \geqslant \alpha\}) = 0\}$$

Define $L^\infty := \{f : f \text{ is a function that is bounded almost everywhere }\}$

Definition 27. We also define $\ell^p(X)$ as $L^p(X)$ with the counting measure for μ. In particular, for $X = \mathbb{N}$, ℓ^p is the collection of complex-valued sequence $\{a_n\}_{n \in \mathbb{N}}$ such that $\sum_{i=1}^{\infty} (a_n)^p < \infty$.

Now, here is the collection of results.

Theorem 25 (The functional analysis)**.** *For $1 \leqslant p \leqslant \infty$ with $1/p + 1/q = 1$, L^p is a Banach space. The norm is given by*

$$\|f\|_p = \begin{cases} \left(\int_X |f|^p \, d\mu\right)^{\frac{1}{p}} & \text{if } p < \infty \\ \operatorname{ess\,sup} |f| & \text{if } p = \infty \end{cases}$$

Theorem 26 (The "Basis")**.** *For $1 \leqslant p < \infty$, the set of simple functions with finite measure, i.e. the set:*

$$\left\{ f : f = \sum_{i=1}^{n} a_i \chi_{E_i}; \mu(E_j) < \infty \text{ for all } 1 \leqslant j \leqslant n, n \in \mathbb{N} \right\}$$

Is dense in $(L^p, \|.\|_p)$.

Theorem 27 (The Inclusion, I)**.** *Let $1 < p < q < r < \infty$, then:*

- *$L^p \cap L^r$ have norm $\|f\| := \|f\|_p + \|f\|_r$, we have $L^q \subset L^p + L^r$, and $\|f\|_p + \|f\|_r \geqslant c\|f\|_q$ for some constant c.*
- *$L^p + L^r$ have norm $\|f\| = \inf\{\|g\|_p + \|h\| + r : f \equiv g + h\}$, with $L^q \subset L^p + L^r$. (Also, we have generalized Hölder as in Theorem 31.)*

Theorem 28 (The Inclusion, II)**.** *If $0 < p < q \leqslant \infty$, then $L^p \subset L^q \Leftrightarrow X$ contains **no** set of arbitrarily small measure, $L^q \subset L^p \Leftrightarrow X$ contains **no** sets of arbitrarily large measure.*

Theorem 29 (The Dual)**.** *Here is a linear map $\psi : L^q \to (L^p)^*$:*

$$\psi(g) = \textit{The } \textbf{\textit{map}} \ \phi_g : L^p \to \mathbb{R}, \qquad \phi_g(f) = \int_X fg \, d\mu.$$

*If $1 \leqslant q < \infty$, **or** $q = \infty$ and μ is semifinite, then ψ is an isometric isomorphism (i.e. $L^q \cong (L^p)^*$).*

That means we really can't say anything about $(L^\infty)^*$. It is quite a messed-up space.

Now let's look at what is the major part of the proof. The "$\|\cdot\|_p$ is indeed a norm" part follows from:

Theorem 30 (Hölder's Inequality)**.** *Let $1 \leqslant p < \infty$, with $1/p + 1/q = 1$, then if $f \in L^p, g \in L^q$, then $fg \in L^1$, and*

$$\|fg\|_1 = \int_X fg \, d\mu \leqslant \|f\|_p \|g\|_q$$

with equality if and only if the following condition holds:

- *$1 < p < \infty$ and $|f(x)|^p = |g(x)|^q$ for (a.e.) $x \in X$, or;*

- $p = 1$ *and* $|g(x)| = \|g\|_\infty$ *on* $\{x \in X : f(x) \neq 0\}$.

Also, when it comes to proving convergence in $\|.\|_p$, it is sometimes beneficial to estimate $|f - f_n| \leqslant 2f$ to apply DCT, but Hölder usually gets the job done equally well.

Theorem 31 (Generalized Hölder)**.** *Let* $0 < p < q < r \leqslant \infty$*, then define* $\lambda = \dfrac{q^{-1} - r^{-1}}{p^{-1} - r^{-1}} \in (0, 1)$*, then* $\|f\|_q \leqslant \|f\|_p^{\lambda} \|f\|_r^{1-\lambda}$.

Theorem 32 (Generalized Hölder, II)**.** *Suppose that* $1 \leqslant p_j \leqslant \infty$ *and* $\sum_1^n p_j^{-1} = r^{-1} \leqslant 1$*. If* $f_j \in L^{p_j}$ *for* $j = 1, \ldots, n$*, then* $\prod_1^n f_j \in L^r$ *and* $\left\|\prod_1^n f_j\right\|_r \leqslant \prod_1^n \|f_j\|_{p_j}$.

Also theorem 13 in Analysis A is useful, let's state it here:

Theorem 33. *If* $f : X \to \mathbb{C}$ *is measurable, there is a sequence* $\{\phi_n\}$ *of simple functions such that*

- $0 \leqslant |\phi_1| \leqslant |\phi_2| \leqslant \cdots \leqslant |f|$*, and*
- $\phi_n \to f$ *pointwise.*

The Inclusion mainly follows from the idea of the A"large-small" decomposition of functions:

Theorem 34. *Assume that* a, b *and* c *are such that* $1 \leqslant a < b < c \leqslant \infty$*. If* $f \in L^b(\mu)$*, show that* f *can be decomposed into a sum of two measurable functions,* $f = g + h$ *such that* $\|g\|_a < \infty$ *and* $\|h\|_c < \infty$.

The important idea of the proof is to consider $E = \{x : |f(x)| < 1\}$ and decompose $f = f\chi_E + f\chi_{X \setminus E}$.
Now, injectivity and isometric part of the dual is easy, the surjective part comes from this huge theorem:

Theorem 35. *Let* $1 \leqslant p \leqslant \infty$*, with* $1/p + 1/q = 1$*, if* g *is measurable, and consider* Γ *to be the set of all functions* f *that is simple, vanish outside a set of finite measure, and* $\|f\|_p = 1$*, then assume the following:*

- $fg \in L^1$ *for all* $f \in \Gamma$*, and;*
- *Either* $\operatorname{supp}(g)$ *is* σ*-finite or* μ *is semifinite.*

Then, define the q***-mass*** *of* g *as*

$$M_q(g) = \sup_{f \in \Gamma} \left| \int_X fg \, d\mu \right|$$

Then $\|g\|_q = M_q(g)$*. In particular, if* $M_q(g) < \infty$*, then* $g \in L^q$.

Note that for those kind of theorem, you always want to do a two-sided "$\leqslant, \geqslant$" proof. For example, $M_q(g) \leqslant \|g\|_q$ is immediate from Hölder's inequality (really? Anything about escaping the supremum that i miss?)
(Small remark. From this theorem to the actual "dual" theorem above needs lots of work, at one point the Radon-Nikodym derivative comes to play again... I don't really understand the details of the proof to be honest.)
Now some useful inequalities:

Theorem 36 (Chebyeshev Inequality)**.** $f \in L^p, 1 \leqslant p < \infty$*, then :*

$$\mu\left(\{x : |f(x)| \geqslant \alpha\}\right) \leqslant \frac{\|f\|_p^p}{\alpha^p}.$$

Theorem 37 (Minkowski Inequality)**.** *Let* $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ *be* σ*-finite measure space, let* $1 \leqslant p < \infty$.

- *if* k *is* $\mathcal{M} \otimes \mathcal{N}$*-measurable, and there is* C *such that*

$$\int_Y |k(x, y)| \, d\nu(y) \leqslant C \text{ for } \mu \text{ a.e. } x, \qquad \int_X |k(x, y)| \, d\mu(x) \leqslant C \text{ for } \nu \text{ a.e. } y$$

Then Consider, if $f \in L^p(\nu)$*, the linear function* $T : L^p(\nu) \to L^p(\mu)$

$$f \longrightarrow Tf(x) := \int_Y k(x, y) f(y) \, d\nu(y)$$

is well-defined (μ*-a.e.), and* $\|Tf\|_{L^p(\mu)} \leqslant C\|f\|_{L^p(\nu)}$.

- *If $f(x,y) \geqslant 0$ for (a.e.) all x, y, then*

$$\left(\left(\int f(x,y)\,dy\right)^p dx\right)^{1/p} \leqslant \left(\left(\int f(x,y)^p\,dy\right)^{1/p} dx\right)$$

The next theorem is super useful to "absorb" the p-norm for μ-a.e. x:

Theorem 38. *If $1 \leqslant p \leqslant \infty$, and the two conditions are satisfied:*

- *$f(\cdot, y) \in L^p(\mu)$ for a.e. y, and;*
- *the function $y \mapsto \|f(\cdot,y)\|_p$ is in $L^1(\nu)$,*

then $f(x,\cdot) \in L^1(\nu)$ for a.e. x, then the function $x \mapsto \int f(x,y)d\nu(y)$ is in $L^p(\mu)$, and

$$\left\|\int f(\cdot,y)d\nu(y)\right\|_p \leqslant \int \|f(\cdot,y)\|_p d\nu(y).$$

Note: The $\cdot$ always mean the quantity is a function of that argument, e.g. $a \cdot$ at the x-component, means it is a fucnction of x.

7 Hilbert Spaces

Definition 28. An **inner product space** (or, an Pre-Hilbert space) is a vector space $(V, \langle\cdot\rangle)$ over $\mathbb{C}$ that is equipped with an inner product. Note, it is a non-degenerate sesquilinear form that is linear on the first variable.

A natural norm (hence a topological structure) is of course $\|x\| = \sqrt{x \bullet x}$. We will use $\bullet$ to represent the dot product, just because I am classic.

Definition 29. A **Hilbert space** is a complete pre-Hilbert space with respect to the norm.

Theorem 39 (Cauchy-Schwartz)**.** *$|x \bullet y| \leqslant \|x\|\|y\|$, with equality iff $x = \lambda y$ for some $\lambda \in \mathbb{C}$.*

Also, $L^2(X)$ is a prime example of Hilbert space (in fact, L^p is not a Hilbert space otherwise), there is a shortcut to show something is *not* a Hilbert space:

Theorem 40. *Let X be a Banach space with norm $\|\cdot\|$. Suppose For all $x, y \in X$, we have the **parallelogram identity**:*

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2).$$

, then

$$\langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2\right)$$

defines an inner product on X. X thus is an Hilbert space $\Leftrightarrow$ the parallelogram identity is satisfied.

Now a technicality, it is useful to synergize this with other convergence:

Theorem 41. *In a Hilbert space, if $\{x_n\} \to x, \{y_n\} \to y$ are convergent sequence, then $\langle x_n, y_n\rangle \to \langle x, y\rangle$ in the normal $\mathbb{C}$ topology.*

Now, there are definition of orthogonality and the orthogonal complement $M^\perp$, but this is normal linear algebra, and I won't repeat the definition here.

Theorem 42 (Pythagoras)**.** *If $\{x_i\}$ are a finite collection of mutually orthogonal vectors then: $\|\sum_i x_i\|^2 = \sum_i \|x_i\|^2$.*

Theorem 43 (Decomposition Theorem)**.** *Given an Hilbert space H and a closed subspace M, we must have $H = M \oplus M^\perp$. Moreover, such a decomposition is unique as $x = y + z$, then $\|x-y\|, \|x-z\|$ is minimal among all such combinations.*

Corollary (Identifying the Dual)**.** *if $f \in H^*$, then there is $y \in H$ such that $f(x) = \langle x, y\rangle$.*

So $H \cong H^*$ in a nautral isomorphism.

The next topic that is important is existence of an **orthonormal basis**. The existences needs Axiom of Choice, but since we don't have the idea of span anymore, we need to replace it with the idea of convergence.

Theorem 44 (Bessel's Inequality)**.** *If $\{u_\alpha\}_{\alpha\in A}$ is an orthonormal set in $\mathcal{H}$, then for any $x \in \mathcal{H}$,*

$$\sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2 \leqslant \|x\|^2.$$

In particular, $\{\alpha : \langle x, u_\alpha\rangle \neq 0\}$ is countable.

Key identity: $\|x+y\|^2 = \|x\|^2 + 2\mathrm{Re}\langle x, y\rangle + \|y\|^2$.

Theorem 45. *If $\{u_\alpha\}_{\alpha\in A}$ is an orthonormal set in $\mathcal{H}$, the following are equivalent:*

- *(Completeness) If $\langle x, u_\alpha\rangle = 0$ for all α, then $x = 0$.*
- *(Parseval's Identity) $\|x\|^2 = \sum_{\alpha\in A} |\langle x, u_\alpha\rangle|^2$ for all $x \in \mathcal{H}$.*
- *For each $x \in \mathcal{H}, x = \sum_{\alpha\in A} \langle x, u_\alpha\rangle u_\alpha$, where the sum on the right has only countably many nonzero terms and converges in the norm topology no matter how these terms are ordered.*

Definition 30. We will call an orthonormal set that satisfies the above condition an **orthonormal basis** of H.

Theorem 46. *There are a few consequences: all H have an orthonormal basis by Zorn's Lemma, and $\{u_\alpha\}_{\alpha\in A}$:*

- *H is separable iff it has an A that is countable. (Recall: Gram-Schimidt process can turn an countable orthogonal set to an orthonormal basis.*
- *There is a unitary map, i.e. a map that preserves the inner product , from $H \to \ell^2(A)$*

$$x \mapsto \widehat{x}, \qquad \widehat{x}_\alpha = \langle x, u_\alpha\rangle$$

8 Convolution and Basic Fourier Analysis

Now our domain is focused on $\mathbb{R}^n$ (or subsets of it, actually), the function we will look at is translation of functions, denoted $\tau_y f(x) := f(x-y)$.

Theorem 47. *Some more elementary fact on $\mathbb{R}^n \to \mathbb{R}$ functions:*

- *If $f \in C_c(\mathbb{R}^n)$, then f is uniformly continuous.*
- *(Translations are continuous in L^p.) If $1 \leqslant p < \infty$, then for any $f \in L^p$,*

$$\|\tau_{y+z} f - \tau_y f\|_p \to 0 \qquad \text{as } z \to 0$$

Definition 31. The **convolution** of f, g is $f * g := \int f(x-y)g(y)\,dy$

The main property of convolution is that it combines two function to form a "nicer" one.

Theorem 48 (Properties of Convolution)**.** *Convolution is associative and commutative.*
Also, For $z \in \mathbb{R}^n, \tau_z(f*g) = (\tau_z f) * g = f * (\tau_z g)$
If A is the closure of $\{x+y : x \in \mathrm{supp}(f), y \in \mathrm{supp}(g)\}$, then $\mathrm{supp}(f*g) \subset A$.

We can apply Minkowski's inequality and get

Theorem 49 (Young's Inequality, three versions)**.**

- *If $f \in L^1, g \in L^p$, $1 \leqslant p \leqslant \infty$,then $f * g$ exists a.e., and $\|f*g\|_p \leqslant \|f\|_1\|g\|_p$.*
- *(Generalized version.) If $1 \leqslant \alpha, \beta, \gamma \leqslant \infty, 1/\alpha + 1\beta = 1 + 1\gamma, f \in L^\alpha, g \in L^\beta$, then $\|f*g\|_\gamma \leqslant \|f\|_\alpha\|g\|_\beta$.*
- *If $f \in L^p, g \in L^q, 1/p + 1/q = 1$, then $f * g \in C_0(\mathbb{R}^n)$, and further, if $f * g \in L^\infty$, then it is continuous.*

The next section is motivated by the idea "L^p convolve with nice function gives nice functions". Firstly, let's look at there indeed exists some nice function.

Theorem 50. *There is $f \in C_c^\infty(\mathbb{R}^n)$.*

Proof. The key is the function $f(x)\exp(-1/x)$, cutoff at $x > 0$ is $C^\infty(\mathbb{R})$, then, to support, say $[0,1]$, we can define $g(x) = f(x)f(1-x)$.

The required function is $f(1 - x_1^2 - \cdots x_n^2)$. □

Now we define $\phi_t(x) = \dfrac{1}{t^n}\phi(tx)$, essentially shrinking the support on a scale of t.

Theorem 51. *Suppose $\phi \in L^1$ and $\int \phi(x)dx = a$. If:*

- *$f \in L^p(1 \leqslant p < \infty)$, then $f * \phi_t \to af$ in the L^p norm as $t \to 0$.*
- *f is bounded and uniformly continuous, then $f * \phi_t \to af$ uniformly as $t \to 0$*
- *$f \in L^\infty$ and f is continuous on an open set U, then $f * \phi_t \to af$ uniformly on compact subsets of U as $t \to 0$.*

A key idea is to get $x - y \cdots dy$ and define the coordinate change $y = tz$ to get ϕ_t to ϕ. For the first bullet point use the "shrink norm" version of Minkowski's identity.

Corollary. *C_c^∞ is dense in L^p and C_0 (with the uniform norm)*

Now we move on to Fourier transform. We start by assuming a function f is $\mathbb{Z}^n$-periodic. A natural thing to look at, is of course the torus $\mathbb{T}^n$. Note that $m(\mathbb{T}^n) = 1$, this is useful for some finiteness estimates. Moreover, we want to use stone-Weierstrass and completeness to produce continuous estimates from $L^2(\mathbb{T}^n)$. The big idea is as follows:

Definition 32. First, we define $E_k(x) = \exp(2\pi ikx)$ for $k \in \mathbb{Z}^n$, then **trigonometric polynomials** are finite linear combinations of $\{E_k\}_{k\in\mathbb{Z}^n}$.

Theorem 52. *$\{E_\lambda\}_{\lambda\in\mathbb{Z}^n}$ is an orthonormal basis for $L^2(\mathbb{T}^n)$.*

Moreover, a check on the Stone-Weierstrass theorem gives us the trigonometric polynomial are dense in $C(\mathbb{T}^n)$, and $C(\mathbb{T}^n)$ are dense in $L^2(\mathbb{T}^n)$ by Hölder with finite measure estimate. Hence,

Theorem 53. *Trigonometric polynomials are dense in $L^2(\mathbb{T}^n)$.*

Definition 33. The above argument shows $\sum_{\lambda\in\mathbb{Z}^n}\langle f, E_\lambda\rangle E_\lambda \to f$ in L^2, this is the **Fourier Series** of f. By the unitary map theorem, this also produces a map $L^2(\mathbb{T}^n) \to \ell^2(\mathbb{Z}^n), f \to \widehat{f}, \widehat{f}_\lambda = \langle f, E_\lambda\rangle$. $\widehat{f}$ is referred as the **Fourier transform** of f.

Theorem 54 (Parseval's Identity)**.** *From the unitary map theorem, we immediately have fourier transform is an isometry, in particular $\|\widehat{f}\|_2 = \|f\|_2$:*

$$\sum_{\lambda\in\mathbb{Z}^n} |\widehat{f}_\lambda|^2 = \int_{\mathbb{T}^n} |f|^2\, dx.$$